# V07: dual isolation valves, redundant DC cooling and IT derating

Executed MATLAB/Simulink R2026a | 8 cases × 168 h | synthetic input data

## 1. Scope and system topology

This is an independent new model preserving the five-state Stateflow dispatcher originating in v04. Two series isolation valves and two independent reject-cooling channels are represented, plus assumed emergency IT thermal-load reduction. The original v85 is unchanged and is NOT the active closed-loop controller. No physical measurements or certified safety are claimed.

## 2. Thermodynamic conservation

The original DC source heat balance closes at every step. Leakage through BOTH damaged isolation valves is subtracted from commanded reject heat rather than invented. Heat not removed by the coolers is explicitly included in the source balance.

C_water = 997 × 500 × 4182 = 2 084 727 000 J/K

T_water(k+1) = T_water(k) + 900 (Q_pool_physical - Q_loss)/C_water

Q_leak = (valve1_stuck AND valve2_stuck AND heat_command=0) * Q_leak_limit

Q_leak_limit = min(25 kW, Q_reject_cmd, 4.5 kW/K * max(0,T_DC-T_pool-3 K))

Q_reject_A = min(Q_reject_remaining, 180 kW * availability_A)

Q_reject_B = min(Q_reject_remaining-Q_A, 150 kW * availability_B)

Q_DC = Q_HX_physical + Q_HP_source + Q_A + Q_B + Q_unhandled

Q_HP_out = Q_HP_source + P_compressor

Q_DC_effective = Q_DC_raw - Q_IT_avoided

Q_IT_avoided = 70% Q_DC_raw when both coolers fail and T_IT_model >= 38 C

T_IT_model(k+1) = max(30, T_IT + 900 (Q_unhandled-Q_passive)/8e6)

## 3. Cooling failover and measurement limits

Two virtual temperature sensors use 15/30-minute delays; a shared -0.4 K bias remained undetected. Main and backup coolers have hypothesized 180 and 150 kW ratings. During their joint failure, automatic derating removes 70% of available modeled DC waste heat when the synthetic IT buffer exceeds 38°C. This buffer temperature is NOT actual server temperature.

## 4. Eight actually executed test results

| Scenario | Leak MWh | Unrej. MWh | Backup MWh | IT max °C |
|---|---|---|---|---|
| Healthy | 0.000 | 0.000 | 0.000 | 30.00 |
| Primary valve stuck | 0.000 | 0.000 | 0.000 | 30.00 |
| Both valves stuck | 0.490 | 0.000 | 0.000 | 30.00 |
| Main REJECT down | 0.000 | 0.000 | 0.080 | 30.00 |
| Both down, no shedding | 0.000 | 0.080 | 0.000 | 47.06 |
| Both down, shedding | 0.000 | 0.050 | 0.000 | 40.33 |
| Common sensor bias | 0.000 | 0.000 | 0.000 | 30.00 |
| Combined failure | 0.146 | 0.039 | 0.000 | 39.26 |

## 5. Independent QA and outstanding work

129251 assertions passed independently over saved CSV: conservation of water, HP and DC heat, two valves, two cooling channels, sensor delay, latch state and derating accounting. Real hardware separation, hydraulic data, common-power faults, IT reaction latency, manufacturer COP and the applicable regulatory edition remain unverified.

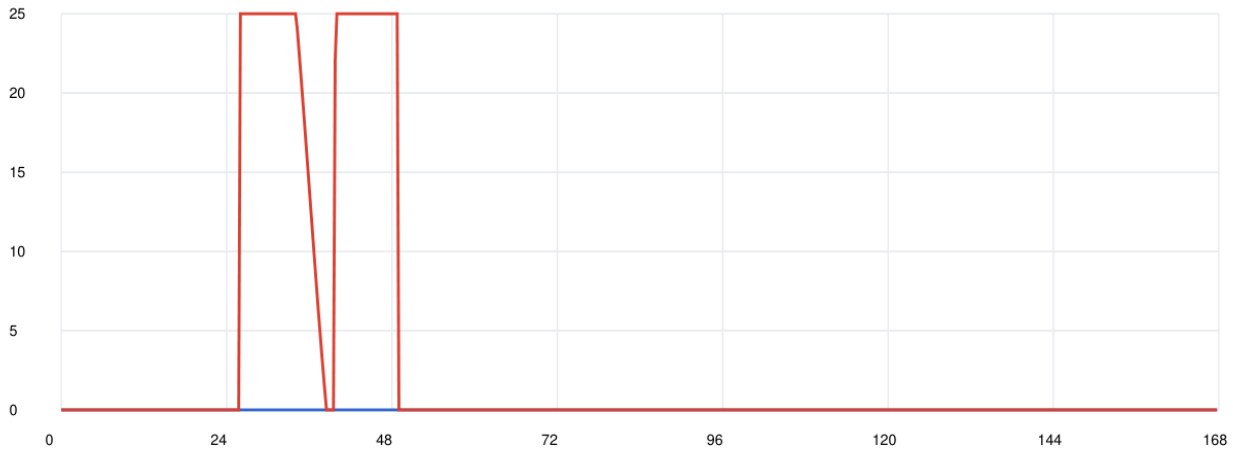
Comparison: unhandled heat 0.07966 MWh -> 0.04989 MWh (approximately 37% reduction).

Source class D = synthetic; main REJECT 180 kW, backup 150 kW are assumptions.
EU Delegated Regulation 2024/1364: https://eur-lex.europa.eu/eli/reg_del/2024/1364/oj

## 6. Plots of executed simulations

Redundancia uzatvaracich ventilov - fyzicke teplo do bazena

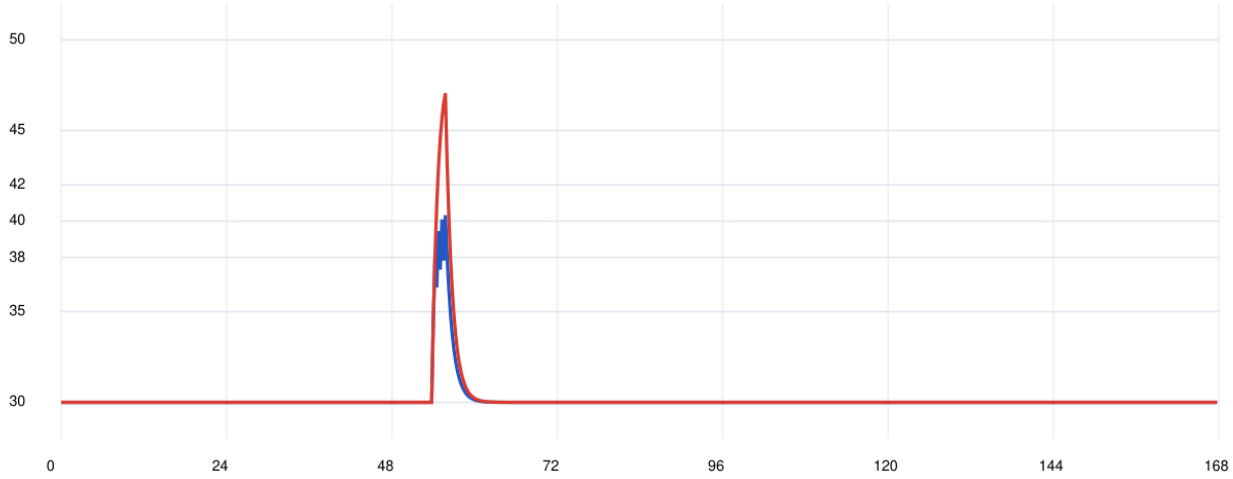
Modra: iba prvy zaseknuty; cervena: oba zaseknute (Q_pool software vypnuty)




Cas [h], os Y = kW fyzickeho uniku

Obe vetvy REJECT mimo prevadzky - efekt znizenia IT zataze

Modra: automaticke odlahcenie, cervena: bez odlahcenia




Os Y: teplota modeloveho IT buffera [C], NIE teplota CPU / serverov.